

ECON 100B — Midterm Study Guide

Classes 1–11 · Jones Chapters 1–6 · Prof. Ryan D. Edwards · Spring 2026

Midterm: March 5, 2026 · Pimentel Auditorium · 60 minutes · 1 page front+back notes allowed

0. Exam Format & Priority Map

Based on the Spring 2014 past midterm (same professor, same models), here is how the exam is structured:

Part	Points	Content	Priority
Whimsical Warmup	~5 pts	Fun multiple-choice — no wrong answers. Free points.	Free
Short Answer Potpourri	~10 pts	Growth rate rules, GDP identities, compound growth, MPK/MPL	HIGH
Applied Solow Question	~25 pts	Draw diagram, comparative statics, short/long-run effects on Y/L	CRITICAL
Applied Romer Question	~25 pts	Derive $g[A]$ and y_t , comparative statics, nonrival goods argument	CRITICAL

Exam Tip

Solow + Romer together $\approx$ 50 pts (two-thirds of the exam). Master diagrams and derivations first.
Short-answer tests growth rate algebra in almost every exam. Know all three rules cold.
Edwards's exams always frame a real-world story (famine, immigration, zombies). Learn to map a story to a parameter change.

Topic	Depth	Likely Exam Role
Ch. 1–2: GDP Accounting	Concise	Short answer: $Y=C+I+G+NX$, real vs nominal
Ch. 3: Growth Math	Thorough	Short answer: compute $g[Y]$, $g[MPK]$, compound growth
Ch. 4–5: Cobb-Douglas & MPK/MPL	Thorough	Short answer + part of applied question
Ch. 6–8: Solow Model	Deep — master all	Full ~25-pt applied question
Ch. 9–11: Romer Model	Deep — master all	Full ~25-pt applied question

1. GDP Accounting (Ch. 1–2) [Concise]

1.1 Expenditure Identity

$$Y = C + I + G + NX$$

C = consumption (~70% of U.S. GDP) ← largest component

I = investment (~15% of U.S. GDP) ← creates new physical capital

G = government (~20% of U.S. GDP)

NX = net exports (~−5% of U.S. GDP) ← U.S. imports more than it exports

Plain English

Everything produced must be bought by someone: households (C), firms (I), government (G), or foreigners net of imports (NX). With the U.S. shares above: $70 + 15 + 20 - 5 = 100\%$.

1.2 Real vs. Nominal GDP

$$Y_N = P \cdot Y_R$$

$$\Rightarrow g[Y_R] = g[Y_N] - \pi$$

P = price level, π = inflation rate = $g[P]$

Example: $g[Y_N] = 5\%$, inflation = $2\% \rightarrow g[Y_R] = 3\%$

Plain English

Nominal GDP rises when prices rise even if nothing real changes. Real GDP removes inflation so you can see actual output growth. To convert: real growth = nominal growth minus inflation.

1.3 PPP (Purchasing Power Parity)

PPP adjusts for price level differences across countries. A dollar buys more in a poor country, so raw exchange-rate comparisons understate poor countries' real income. Use PPP-adjusted figures to compare living standards internationally.

2. Growth Math (Ch. 3) [HIGH PRIORITY — Short Answer Staple]

Growth rate algebra is tested in almost every short-answer section. The three rules below must be automatic.

2.1 Compound Growth

$$x_t = x_0 \cdot (1 + g)^t$$

x_0 = starting value, g = growth rate per period, t = number of periods

Example: \$100 at $g = 3\%$ for 10 years $\rightarrow 100 \times (1.03)^{10} \approx \134

Plain English

Like compound interest: each year you grow on top of the previous year's growth. Even a small g compounds to a large multiplier over many periods.

2.2 The Three Growth Rate Rules ★ Must Be Automatic

$$\text{Rule 1 (Product): } g[X \cdot Y] = g[X] + g[Y]$$

$$\text{Rule 2 (Ratio): } g[X / Y] = g[X] - g[Y]$$

$$\text{Rule 3 (Power): } g[X^a] = a \cdot g[X]$$

Any constant (like $\frac{1}{3}$, $\bar{A}$, $\bar{s}$, $\bar{d}$) has growth rate = 0.

Plain English

Rule 1: If you multiply two things, their growth rates ADD. Example: Nominal GDP = $P \times$ Real GDP, so $g[\text{Nom}] = \pi + g[\text{Real}]$.

Rule 2: If you divide, SUBTRACT growth rates. Example: Real GDP = Nom/P , so $g[\text{Real}] = g[\text{Nom}] - \pi$.

Rule 3: If something is raised to a power a , multiply the growth rate by a . Example: $g[K^{1/3}] = \frac{1}{3} \cdot g[K]$.

Key Application 1: Growth Rate of Cobb-Douglas Output

$$g[Y] = g[\bar{A}] + \frac{1}{3} \cdot g[K] + \frac{2}{3} \cdot g[L]$$

Apply Rules 1 and 3 to $Y = \bar{A} \cdot K^{1/3} \cdot L^{2/3}$. Each factor contributes (exponent $\times$ growth rate).

Key Application 2: Growth Rate of the Wage

$$\text{MPL} = \frac{2}{3} \cdot \bar{A} \cdot (K/L)^{1/3}$$

$$g[\text{MPL}] = g[\bar{A}] + \frac{1}{3} \cdot (g[K] - g[L])$$

If $\bar{A}$ rises 5% and K/L is constant: $g[\text{MPL}] = 0.05 \rightarrow$ wages rise exactly 5%.

2.3 Rule of 70

$$\text{Doubling time} \approx 70 / (\text{growth rate in \%})$$

$g = 2\%/yr \rightarrow$ doubles every ~ 35 years $g = 7\%/yr \rightarrow$ doubles every ~ 10 years

2.4 Ratio (Log) Scale

On a log scale, a **straight line = constant growth rate**. The slope of the line equals the growth rate. Used in the Romer time-path diagram: parallel lines = equal growth, steeper slope = faster growth.

3. Model of Production — Cobb-Douglas (Ch. 4–5) [MEDIUM-HIGH PRIORITY]

Foundation for both Solow and Romer. Know how to derive MPK and MPL, apply growth rate rules to them, and analyze comparative statics.

3.1 Cobb-Douglas Production Function

$$Y = \bar{A} \cdot K^{1/3} \cdot L^{2/3}$$

Y = output/GDP

$\bar{A}$ = Total Factor Productivity (technology, institutions, management quality)

K = physical capital stock (machines, buildings, infrastructure)

L = labor (number of workers)

$1/3$ = capital's income share | $2/3$ = labor's income share

Plain English

Output is a function of how productive the technology is ($\bar{A}$), how much physical capital there is (K), and how many workers (L). The exponents $1/3$ and $2/3$ are calibrated to U.S. data: workers earn $\sim 2/3$ of national income; capital owners earn $\sim 1/3$.

3.2 Constant Returns to Scale (CRS)

$$F(\bar{A}, \lambda K, \lambda L) = \lambda \cdot Y \quad \text{for any } \lambda > 0$$

Proof: $\bar{A} \cdot (\lambda K)^{1/3} \cdot (\lambda L)^{2/3} = \bar{A} \cdot \lambda^{1/3+2/3} \cdot K^{1/3} \cdot L^{2/3} = \lambda \cdot Y \quad \checkmark$
(exponents sum to 1: $1/3 + 2/3 = 1 \rightarrow$ the λ terms combine to λ^1)

Plain English

Double ALL inputs (K and L together) $\rightarrow$ output exactly doubles. This is about scaling the whole economy, not about adding just one input. You can have CRS overall and still have diminishing returns to each individual input.

3.3 Per-Worker Form ★ Used Throughout Solow

$$y = \bar{A} \cdot k^{1/3} \quad k = K/L \text{ (capital per worker, or capital-labor ratio)}$$

Derivation: divide $Y = \bar{A} \cdot K^{1/3} \cdot L^{2/3}$ by L on both sides:

$$Y/L = \bar{A} \cdot (K/L)^{1/3} = \bar{A} \cdot k^{1/3}$$

Plain English

Instead of tracking total K and total L separately, the Solow model tracks how much capital each worker has (k). Output per worker (y) is what determines living standards.

3.4 Marginal Products and Factor Prices ★ Know These

$$\text{MPK} = 1/3 \cdot \bar{A} \cdot (L/K)^{2/3} = 1/3 \cdot Y/K = r \text{ (interest rate)}$$

$$MPL = \frac{2}{3} \cdot \bar{A} \cdot (K/L)^{\frac{1}{3}} = \frac{2}{3} \cdot Y/L = w \text{ (wage)}$$

In competitive markets each factor is paid its marginal product.

Capital's total income: $r \cdot K = \frac{1}{3} \cdot Y$ Labor's total income: $w \cdot L = \frac{2}{3} \cdot Y$

Plain English

MPK: how much extra output you get from one more unit of capital (with L fixed). Capital gets paid this in competitive markets → it equals the real interest rate r.

MPL: how much extra output one more worker adds → it equals the wage w.

Diminishing returns: as K rises (L fixed), MPK falls. Each additional machine adds less output because workers can only operate so many machines. This eventually stops capital accumulation in the Solow model.

3.5 Growth Rates of Factor Prices

$$g[\text{MPK}] = g[\bar{A}] - \frac{2}{3} \cdot g[k]$$

$$g[\text{MPL}] = g[\bar{A}] + \frac{1}{3} \cdot g[k]$$

where $k = K/L$. If $\bar{A}$ constant and k constant: $g[\text{MPK}] = g[\text{MPL}] = 0$

If K rises faster than L ($g[k] > 0$): MPK falls, MPL rises.

3.6 Immigration Effects in the Production Model ★ Exam Staple

If L suddenly rises (immigration) while K stays fixed:

$$k = K/L \text{ falls} \Rightarrow \text{MPL} = w \text{ falls (short run)}$$

Total output Y RISES (more workers → more production)

Output per worker $y = Y/L$ FALLS (each worker has less capital)

Plain English

More workers dilute the capital stock. Each worker has less to work with → each is less productive → wages fall in the short run. This is diminishing returns to labor.

Exam Tip

Always distinguish total output Y (goes UP with immigration) from output per worker Y/L (goes DOWN when K is fixed). The exam typically asks about both.

4. Solow Growth Model (Ch. 6–8) [CRITICAL — ~25 pts]

You must be able to: (1) draw and label the diagram from memory, (2) derive the steady state algebraically step by step, (3) apply comparative statics correctly, and (4) analyze real-world scenarios.

4.1 Capital Accumulation — Core Equation

$$\Delta K = sY - dK$$

s = saving/investment rate (fraction of income saved and invested)

d = depreciation rate (fraction of K that wears out each period)

sY = gross investment (new capital created)

dK = capital that depreciates

Plain English

The capital stock grows by new investment (sY) and shrinks by depreciation (dK). In a closed economy, saving = investment. Capital doesn't reproduce itself — you need saving to build it.

4.2 The Solow Diagram ★ Draw This From Memory

Curve	Equation	Shape	Meaning
Income curve	$Y = \bar{A} \cdot K^{1/3} \cdot L^{2/3}$	Concave (like $\sqrt{K}$)	Total output at each K
Saving curve	$sY = s \cdot \bar{A} \cdot K^{1/3} \cdot L^{2/3}$	Same shape, scaled by s below Y	New capital added per period
Depreciation line	dK	Straight through origin, slope = d	Capital worn out per period

$$\text{Steady state } K^*: sY = dK \implies \Delta K = 0$$

At K^* : capital, output, and income per person are all constant.

Below K^* : $sY > dK \rightarrow K$ grows rightward toward K^*

Above K^* : $sY < dK \rightarrow K$ shrinks leftward back to K^*

Exam Tip

Draw: concave Y curve (top), concave sY curve (below Y , parallel shape), straight dK line from origin.

Mark K^* where sY meets dK . Mark Y^* as the height of Y above K^* . Label the intersection clearly.

For comparative statics: draw BEFORE curves (dashed), AFTER curves (solid), show old K^* and new K^{**} with arrows.

4.3 Steady-State Derivation ★ Show Every Step

Step 1 — Set $\Delta K = 0$:

$$s \cdot \bar{A} \cdot K^{1/3} \cdot L^{2/3} = d \cdot K$$

Step 2 — Divide both sides by $K^{1/3}$:

$$s \cdot \bar{A} \cdot L^{2/3} = d \cdot K^{2/3}$$

Step 3 — Raise both sides to the power $3/2$:

$$(s \cdot \bar{A} \cdot L^{2/3})^{3/2} = d^{3/2} \cdot K$$

Step 4 — Solve for K^* :

$$K^* = (s\bar{A}/d)^{3/2} \cdot L$$

Step 5 — Find $y^* = Y^*/L = \bar{A} \cdot k^{*1/3}$ where $k^* = K^*/L = (s\bar{A}/d)^{3/2}$:

$$y^* = \bar{A}^{3/2} \cdot (s/d)^{1/2}$$

$$K^*/Y^* = s/d \quad (\text{the capital-to-output ratio at steady state})$$

Plain English

At steady state, new investment exactly replaces what wears out. Everything stops changing.

y^* depends on $\bar{A}$, s , d — but NOT on L . Population size does not affect long-run living standards in Solow.

Example: $s = 20\%$, $d = 5\% \rightarrow K^*/Y^* = 4$. The U.S. capital-to-GDP ratio is roughly 3–4.

4.4 Transition Dynamics

$K < K^*$: $sY > dK \rightarrow \Delta K > 0 \rightarrow K$ grows toward K^* (fast when far below)

$K > K^*$: $sY < dK \rightarrow \Delta K < 0 \rightarrow K$ falls toward K^* (slow when just above)

Long-run $g[y] = 0$ once at K^* ← this is the key failure of the Solow model

Plain English

The economy converges to K^* like a ball rolling to the bottom of a bowl. Poor countries far below K^* grow fast; rich countries near K^* grow slowly. This is 'conditional convergence.' But growth eventually stops — the model's main weakness.

4.5 Comparative Statics ★ Critical — Exam Always Tests This

For every parameter change: (1) identify which curve shifts, (2) show new K^{**} , (3) describe the short-run and long-run path.

Shock	Curve Change	New K^*	New y^*	Mechanism
↑ s (saving rate)	Saving curve sY shifts UP. dK unchanged.	Higher	Higher	More investment per period → higher steady-state capital
↑ d (depreciation)	Depreciation line dK rotates UP (steeper slope).	Lower	Lower	Capital wears out faster → harder to maintain the stock
↑ $\bar{A}$ (productivity)	BOTH Y and sY shift UP (since $sY = s \cdot Y$). dK unchanged.	Higher	Much higher	More output → more saving → more K^* ; both level and growth path improve
One-time ↑ L (immigration)	BOTH Y and sY shift UP. dK unchanged.	Higher total K^* , but $k^* = K^*/L$ unchanged in LR	y^* unchanged in LR	Short run: $k = K/L$ falls → y falls. Long run: K adjusts, k returns to same k^*

Exam Tip

When $\bar{A}$ changes, you MUST shift BOTH the income curve Y and the saving curve sY — forgetting to shift both is the most common mistake.

The Irish Famine (past midterm Q3): potato blight kills productivity $\rightarrow \bar{A}$ falls $\rightarrow$ BOTH Y and sY shift down $\rightarrow$ lower K^* and y^* .

4.6 Solow with Population Growth ★ Important Variant

When population grows at rate n , capital per worker gets 'diluted' among more workers. Work in per-worker units $k = K/L$:

$$\Delta k = s \cdot \bar{A} \cdot k^{1/2} - (n + d) \cdot k$$

n = population growth rate | $(n+d)$ = effective depreciation on k

Solow diagram: x-axis = k , depreciation line slope = $(n+d)$ not just d

Steady-state k^* and y^* (set $\Delta k = 0$ and solve):

$$k^* = [s\bar{A} / (n+d)]^{2/3}$$

$$y^* = \bar{A}^{1/3} \cdot [s / (n+d)]^{2/3}$$

$$K^*/Y^* = s / (n+d)$$

Plain English

Growing population dilutes capital: new workers arrive without bringing their own machines. The effective 'depreciation' rate on k is now $(n+d)$ instead of just d .

Higher $n \rightarrow$ lower k^* and lower y^* . Countries with faster population growth are predicted to be poorer per person — a prediction supported by cross-country data.

4.7 Immigration Analysis ★ Guaranteed Exam Topic

Scenario A — One-time large increase in L (sudden immigration inflow):

	Short Run (K fixed)	Long Run (K adjusts)
$k = K/L$	Falls immediately (more workers, same K)	Returns to same k^*
$y = Y/L$	Falls (less capital per worker)	Returns to same y^*
Wage $w = MPL$	Falls (lower $k \rightarrow$ lower MPL)	Returns to same w^*
Total output Y	Rises immediately	Rises permanently
Net effect on y^*	Temporary decline	$y^* = \bar{A}^{1/3} \cdot (s/(n+d))^{2/3}$ — unchanged (depends only on $\bar{A}$, s , n , d , not L level)

Scenario B — Permanent reduction in population growth rate $n \rightarrow n' < n$:

	Short Run	Long Run
k	Unchanged immediately	Rises gradually to new higher k^{**}
y	Unchanged immediately	Rises permanently to new higher y^{**}
Why different from Scenario A?	n affects ongoing flow of new workers, not current stock	Lower $n \rightarrow$ flatter depreciation line $\rightarrow$ new higher steady state

Plain English

Scenario A (one-time): Short-run wage pressure, but Solow predicts NO permanent effect on per-person income. Capital adjusts and k returns to k^* .

Scenario B (lower n): Permanently higher y^* . Less ongoing dilution $\rightarrow$ capital per worker ratchets up to a new higher steady state.

Key distinction: one-time L shock affects the stock (temporary), n change affects the ongoing flow (permanent).

4.8 Solow's Main Failure

Key Concept

At steady state, $g[y] = 0$. Living standards stop growing. But the U.S. has grown $\sim 2\%$ per person per year for 150 years.

Solow treats $\bar{A}$ as fixed and unexplained. It cannot account for the empirical fact that TFP keeps rising.

This motivates the Romer model: sustained growth comes from deliberate investment in new ideas.

5. Romer Model of Ideas (Ch. 9–11) [CRITICAL — ~25 pts]

Focus on: (1) the four setup equations, (2) derivation of $g[A]$ and y_t , (3) comparative statics, and (4) why the free market underproduces ideas.

5.1 Rival vs. Nonrival Goods ★ The Central Distinction

Type	Definition	Examples	Simultaneous use?
Rival	Using it prevents others from using it	A tractor, a worker's time, land	No
Nonrival	Using it does NOT prevent others	A formula, software, a recipe, a patent	Yes — unlimited copies at zero cost

Plain English

Capital K and labor L are rival (a machine can only be in one place at once). Technology A — the stock of ideas — is nonrival.

A blueprint for a better process: once known, every firm in the world can use it simultaneously, with no reduction in anyone else's access.

Nonrivalry creates Increasing Returns to Scale: double K and L (rival inputs) AND use the same ideas → output more than doubles. IRS is incompatible with perfect competition → market failure.

5.2 The Four Model Equations ★ Memorize and Understand Each One

Eq 1 (Idea production):

$$\Delta A_t = \bar{z} \cdot A_t \cdot L_{at}$$

Eq 2 (Goods production):

$$Y_t = A_t \cdot L_{yt}$$

Eq 3 (Labor market):

$$\bar{N} = L_{at} + L_{yt}$$

Eq 4 (Scientist allocation):

$$L_{at} = \ell \cdot \bar{N}$$

A_t = stock of ideas at time t

L_{at} = scientists (researchers)

L_{yt} = farmers (goods workers)

$\bar{N}$ = total fixed population

$\bar{z}$ = researcher productivity

ℓ = fraction of population in research (policy parameter)

Plain English

Eq 1: New ideas = (existing knowledge) × (scientists) × (productivity $\bar{z}$). The A_t on the right side captures 'standing on the shoulders of giants' — more existing knowledge makes new discoveries easier.

Eq 2: Output = technology × number of farmers. A_t is a nonrival multiplier: the same ideas raise the productivity of ALL farmers simultaneously.

Eqs 3 & 4: Everyone is either a scientist or a farmer. The fraction ℓ is determined by policy (e.g., R&D subsidies).

5.3 Solving the Romer Model ★ Two-Step Derivation

Step 1 — Find $g[A]$ (divide Eq 1 by A_t):

$$g[A] = \Delta A_t / A_t = \bar{z} \cdot L_{at}$$

Substitute $L_{at} = \ell \cdot \bar{N}$ from Eq 4:

$$g[A] = \bar{z} \cdot \ell \cdot \bar{N} \quad (\text{constant})$$

Step 2 — Find y_t (output per person):

$$A_t = A_0 \cdot (1 + g[A])^t = A_0 \cdot (1 + \bar{z} \cdot \ell \cdot \bar{N})^t$$

$$L_{yt} = (1 - \ell) \cdot \bar{N} \rightarrow Y_t = A_t \cdot (1 - \ell) \cdot \bar{N}$$

$$y_t = Y_t / \bar{N} = A_0 \cdot (1 + \bar{z} \cdot \ell \cdot \bar{N})^t \cdot (1 - \ell)$$

Plain English

$g[A]$ is constant and positive $\rightarrow$ technology and income per person grow FOREVER at a constant rate. This fixes Solow's main problem of growth stopping at steady state.

The $(1 - \ell)$ term: more scientists (higher ℓ) means faster growth but fewer farmers producing goods today. There is an explicit trade-off between the growth rate and the current consumption level.

5.4 Comparative Statics ★ Know All Three

Change	Effect on $g[A]$	Effect on y_t level	Plain English
$\uparrow \bar{z}$ (better research tools)	$\uparrow$ rises	Steeper growth path, same starting level	More ideas per scientist $\rightarrow$ pure growth gain, no trade-off
$\uparrow \ell$ (more scientists)	$\uparrow$ rises	DROPS immediately (fewer farmers), then grows faster	Pay a one-time level cost to get permanently higher growth
$\uparrow \bar{N}$ (larger population)	$\uparrow$ rises	Steeper path, same starting level	More people = more potential scientists = more ideas. Scale effect.

5.5 Time Path Diagram ★ Draw for Exam

On a log scale, y_t grows as a straight upward line with slope $= g[A] = \bar{z} \cdot \ell \cdot \bar{N}$.

Effect of $\uparrow \ell$ at time t_0 (more scientists from that point on):

1. At t_0 : y_t drops (level effect — fewer farmers producing today)
2. After t_0 : y_t grows at a steeper slope (higher $g[A]$)
3. Eventually crosses the old path — higher ℓ is worth it in the long run

Effect of $\uparrow \bar{z}$ at time t_0 :

1. At t_0 : y_t unchanged (no level effect)
2. After t_0 : y_t grows at a steeper slope (pure growth improvement)

5.6 Why the Free Market Underproduces Ideas ★ Key Exam Argument

This is the logic the exam tests in a multi-part question:

1. **Step 1 — Goods sector:** Farmers' MPL = A_t . Total wages paid to farmers = $A_t \cdot L_{yt} = Y_t$. All output goes to workers.
2. **Step 2 — Nothing left:** Subtract farmer wages from output: $Y_t - A_t \cdot L_{yt} = 0$. Zero remains for any other factor.
3. **Step 3 — Scientists can't earn wages:** Scientists' marginal product in ideas = $\bar{z} \cdot A_t$. But ideas are nonrival — anyone can copy them freely. Scientists cannot sell their marginal product for food. Their effective market wage = 0.
4. **Step 4 — Market equilibrium:** No one becomes a scientist (they'd starve). $\ell \rightarrow 0$. With no scientists, $g[A] = \bar{z} \cdot 0 \cdot \bar{N} = 0$. Long-run growth stops. The free market equilibrium has zero growth.

Key Concept

Market failure: ideas are a public good (nonrival + non-excludable). Scientists cannot capture the social value of their discoveries → private market underproduces research.

Government fix: tax goods producers (farmers), pay scientists. The tax rate that equalizes wages is $\tau = \ell$. This was the past midterm Q5 calculation.

5.7 Growth Accounting (Section 6.5)

Rearrange $g[Y] = g[A] + \frac{1}{3} \cdot g[K] + \frac{2}{3} \cdot g[L]$ to isolate TFP growth:

$$g[A] = g[Y] - \frac{1}{3} \cdot g[K] - \frac{2}{3} \cdot g[L] \quad (\text{Solow Residual})$$

Per-worker version:

$$g[Y/L] = g[A] + \frac{1}{3} \cdot g[K/L]$$

In U.S. data: roughly $\frac{1}{3}$ of per-worker growth = capital deepening, $\frac{2}{3}$ = TFP growth.

Plain English

We can't measure A directly, but we can compute its growth rate as what's left over after accounting for K and L contributions. The Romer model explains where this residual comes from: deliberate R&D investment.

6. Solow vs. Romer — Side-by-Side

Feature	Solow (Ch. 6–8)	Romer (Ch. 9–11)
Engine of growth	Capital accumulation: $\Delta K = sY - dK$	Idea accumulation: $\Delta A = \bar{z} \cdot A \cdot L_a$
Long-run $g[y]$	$= 0$ at steady state (stops)	$= g[A] = \bar{z} \cdot \bar{\ell} \cdot \bar{N}$ (permanent, constant)
Returns to scale	CRS in rival inputs K and L	IRS when nonrival ideas A are included
Role of saving rate s	Sets LEVEL of y^* — not the growth rate	Not directly in the model
Role of population N	$\uparrow L \rightarrow \uparrow$ total Y ; no effect on y^*	$\uparrow \bar{N} \rightarrow$ more scientists $\rightarrow$ faster $g[A] \rightarrow$ faster growth
Market failure?	No — competitive markets work	Yes — nonrival ideas $\rightarrow$ scientists earn zero $\rightarrow \bar{\ell} = 0$
Policy implication	Increase s to raise y^*	Subsidize R&D to support $\bar{\ell} > 0$
Convergence?	Yes — conditional convergence	Not guaranteed — scale effects favor large/innovative countries
Nobel connection	Solow 1987	Romer 2018; Acemoglu (institutions) 2024

7. Quick Formula Reference

GDP and Growth Math

$$\begin{aligned}
 Y &= C + I + G + NX \\
 g[Y_R] &= g[Y_N] - \pi \\
 x_t &= x_0 \cdot (1+g)^t \quad \text{Doubling time} \approx 70 / g\% \\
 g[X \cdot Y] &= g[X] + g[Y] \quad g[X/Y] = g[X] - g[Y] \quad g[X^a] = a \cdot g[X]
 \end{aligned}$$

Model of Production

$$\begin{aligned}
 Y &= \bar{A} \cdot K^{1/3} \cdot L^{2/3} \quad y = \bar{A} \cdot k^{1/3} \quad (k = K/L) \\
 MPK &= 1/3 \cdot \bar{A} \cdot (L/K)^{2/3} = r \quad MPL = 2/3 \cdot \bar{A} \cdot (K/L)^{1/3} = w \\
 g[Y] &= g[\bar{A}] + 1/3 \cdot g[K] + 2/3 \cdot g[L]
 \end{aligned}$$

Solow Model

$$\begin{aligned}
 \Delta K &= sY - dK \quad \Delta k = s \cdot \bar{A} \cdot k^{1/3} - (n+d) \cdot k \\
 K^* &= (s\bar{A}/d)^{3/2} \cdot L \quad k^* = [s\bar{A}/(n+d)]^{3/2} \\
 y^* &= \bar{A}^{3/2} \cdot (s/d)^{1/2} \quad y^* = \bar{A}^{3/2} \cdot [s/(n+d)]^{1/2} \\
 K^*/Y^* &= s/d \quad (\text{or } s/(n+d) \text{ with population growth})
 \end{aligned}$$

Romer Model

$$\begin{aligned}
 \Delta A_t &= \bar{z} \cdot A_t \cdot L_{at} \quad Y_t = A_t \cdot L_{yt} \quad L_{at} = \ell \cdot \bar{N} \\
 g[A] &= \bar{z} \cdot \ell \cdot \bar{N} \\
 y_t &= A_0 \cdot (1 + \bar{z} \cdot \ell \cdot \bar{N})^t \cdot (1 - \ell) \\
 g[A] &= g[Y] - 1/3 \cdot g[K] - 2/3 \cdot g[L] \quad (\text{Solow Residual / TFP growth})
 \end{aligned}$$

Exam Skills Checklist

- Growth rules:** Apply $g[XY]=g[X]+g[Y]$ to any formula instantly. No hesitation.
- Solow diagram:** Draw Y (concave), sY (below Y), dK (straight line) from memory. Label K^* , Y^* , intersection.
- Comparative statics:** Identify which curve(s) shift. Show old K^* and new K^{**} with arrows. State short-run and long-run separately.
- Story → parameter:** Famine → $\downarrow \bar{A}$. One-time immigration → $\uparrow L$. Slower pop growth → $\downarrow n$. Scientists become zombies → $\ell \rightarrow 0$.
- Derive K^* :** Set $\Delta K=0$, substitute $Y=\bar{A} \cdot K^{1/3} \cdot L^{2/3}$, solve. Show every algebra step.
- Derive $g[A]$:** Divide ΔA by $A \rightarrow g[A] = \bar{z} \cdot \ell \cdot \bar{N}$. Substitute $L_{at} = \ell \cdot \bar{N} \rightarrow g[A] = \bar{z} \cdot \ell \cdot \bar{N}$.
- Short vs. long run:** Short run: K (or k) is fixed. Long run: K adjusts to new steady state. Always state BOTH.
- Nonrival argument:** Rival K , $L \rightarrow$ CRS $\rightarrow$ competitive markets work. Add nonrival ideas $\rightarrow$ IRS $\rightarrow$ market underproduces research.